

Assignment 1 – AI log

Below is a log of the interactions I had with ChatGPT Edu when working on this assignment.

Prompt(s)	Why?	Incorporated as...
Is it normal that a BPE tokenizer with a given vocabulary size of 2,000 has 0 merges? I'm using HuggingFace's tokenizers library. Do my instructions really mean number of merges when talking about vocabulary size? [+ instruction excerpt] Isn't it weird that my vocabulary ends up being 9,443 characters big even after explicitly setting vocab_size to 2,000? [+ code]	This confused me a lot.	Vocabulary sizes of 10,000 and 20,000 for Small and Big BPE, respectively.
Why aren't there blank spaces in the sample BPE tokenizations? [+ sample tokenizations] What are my options if I skip Whitespace pre-tokenization? I need some kind of pre-tokenization to be able to use tokenizer's from_file loading.	I was confused.	Metaspace pre-tokenization instead of than Whitespace.
Is a character-level tokenizer a tokenizer that literally turns every character into a token? Or is it the same as a unigram tokenizer? What would x and y be in a decoder-only Transformer language model? Would I need special [PAD] and [MASK] tokens for this model? [+ description]	I wanted this to be 100% clear before I started working on it.	A character-level tokenizer. X as a sequence of tokens and y as the same sequence but shifted to the right (AI first suggested only one step to the right but later we agreed that one character-length to the right would be more efficient.). No [PAD] or [MASK] tokens, but instead an [EOS] token.
Is there a more efficient way to load this data from JSON? [+ code] Is there a more efficient way to build these two lists? [+ code] Is there a way to make this token counting more efficient? [+ code]	My code just felt ugly.	No change – apparently my code was as efficient as could be. Moved some code so I could use one loop instead of two. Switched from regular dictionaries to Counters and defaultdicts. Changed the order of the loops.
What's the best way of structuring this data? [+ description]	I wanted my codebase to be	Separate folders for tokenizers, plots, and models.

	structured well.	
Is there a way to format an f-string to get at most x characters in each cell when building a table? Is there a way to center-align the header as well?	I wanted my tables to look pretty.	Pretty tables.
What do B and T refer to here? [+ function description) Can you explain what torch.arange() does?	The official documentation confused me.	B = batch size, T = context length. I used it in my code.
How do I force PyTorch to use the server's GPU? And where in my code do I need to specify this? What if the GPU runs out of memory?	I wasn't familiar with this.	Added a bunch of .to(device) in various places. Reduced batch size from 64 to 32.
Here are the training and validation losses per epoch: [data]. After how many epochs do you think I should stop training?	I wanted to discuss this with someone.	AI suggested 5-10 after some discussion; I went with 8 after trying 3, 4, 5, and 8.
What is the PyTorch method for extracting parameter counts for a model?	I forgot.	Model.parameters (duh)
How do I best combine these three plots into one? [description]	I wanted to create a comparison plot.	The comparison plot in the report (and in plots/loss_comparison.png). I used different colors for different models and then different types of lines for training vs. validation losses.
Is it normal that my data consists of 9 million ids after tokenization? Does it make sense to build my own DecoderTransformer and TransformerBlock classes in PyTorch or are there ready-made blocks available for this? Does this architecture for a Transformer class make sense? [description] Each model needs a separate optimizer when they have different tokenization, right? Is it normal that the loss evaluation takes >5 minutes for this setup? [description]	Sanity checks when coding.	Yes. No change. Built my own classes to have more control of especially the masking. Switched from ELU non-linearity to GELU + added an extra normalization layer after the transformer block + removed dropout from the multi-head attention layer. Yes. No change. Switched from a stride of 1 to context_length to make evaluation more efficient.

Doesn't nats per character (NPC) make more sense than bits per character (BPC) if I'm using cross-entropy loss? Is this logic for calculating token counts per language and shared tokens reasonable? [+ code] Are these BPE token distributions across languages reasonable? [numbers] Do these parameter counts make sense for these models? [numbers + descriptions] Are these NPC values reasonable for these setups? [numbers + descriptions] Is overall NPC just the average of the language-specific NPCs?		Used NPC instead of BPC. Slightly altered the shared tokens calculation to avoid double and triple counting. Yes, apparently. No change. Yes, apparently. No change. Yes, apparently. No change. No, that's macro-average NPC, where each language gets equal weight. Instead, I used micro NPC, where each character gets equal weight.
Here is my justification for keeping epochs constant: [text]. Do you think it is reasonable? What would a "brief explanation of your training-control policy" normally involve? Is this description enough? [+ text] Is this description of nats per character correct? [+ text] Do you agree that Char-level being so competitive with BPE is surprising? Here is my explanation for why the token distribution changed when increasing BPE vocabulary size: [text]. Do you think it is reasonable?	Sanity check when writing.	Hedged the practicality argument. Removed an argument about larger tokens being more meaningful training examples. Toned down some unnecessarily strong language. Yes, apparently. No change. Yes, apparently. No change. AI did agree, but pointed to my experimental setup as a possible reason. I expand upon this in my report. Mostly yes, but my understanding of Chinese was not fully accurate. I updated this in my report.
Why is my saving function creating generator expressions rather than actual tokens/characters? [+ code]	I couldn't figure out the bugs.	Squashed bugs.

Why are Chinese characters turning into Unicode codes instead of actual characters? [+ code] Why does my CharTokenizer become NoneType when loading from a file? [+ code] Why am I getting this error after removing Whitespace pre-tokenization? [+ error] Why do I get KeyError when trying to print the stats? [+ code] I guess I'm getting this error because my CharEncoding class is missing a magic method. Which one if so, and what does it do? [+ error] Why am I getting a TypeError when extracting tokens? [+ code] Why are my plots missing the legend? [+ code] Why is the loss function complaining about matrix size? [+ error] Why do I get this size mismatch error when initializing the model? [+ error]		
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